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Portable Coordinate Measurement Arm Joint Design

Project Proposal

LN Matota

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2026

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Declaration

- ☐ I have discussed the use (or non-use) of GenAI tools with my supervisor, and we have agreed on appropriate boundaries.
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Institutional Acknowledgement

This declaration aligns with Stellenbosch University's Position Statement on the ethical use of AI in Research and Teaching-Learning-Assessment¹, the Draft interim SU guidelines on allowable AI use and academic integrity in assessment², and draws on international best practices.

¹Position Statement on Ethical use of Artificial Intelligence in Research and Teaching-Learning Assessment

²Draft interim SU guidelines on allowable AI use and academic integrity in assessment

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List of Abbreviations

3D 3D

AACMM Articulated Arm Coordinate Measuring Machine

CAD Computer-Aided Design

CMM Coordinate Measurement Machine

D-H Denavit-Hartenberg

FEA Finite-Element Analysis

FIA first article inspection

PCMA Portable Coordinate Measurement Arm

SPAT Single Point Articulation Test

List of Symbols

1 Introduction

1.1 Background

A Portable Coordinate Measurement Arm (PCMA) is a handheld metrology instrument that captures precise 3D (3D) coordinates directly from a physical object [1]. A probe is used to take measurements of an object that are plotted on a 3D plane as coordinates to create a working model [1].



Figure 1: A Portable Coordinate Measurement Arm

PCMAAs can also be referred to as an Articulated Arm Coordinate Measuring Machine (AACMM)³ in literature, offer flexible metrology for multiple applications, such as dimensional analysis, Computer-Aided Design (CAD) inspection, reverse engineering etc [3]. However, these machines have rely on “human” actuation that produces a large amount of uncertainty in the measurements taken. The main limitation of these joints is the “human factor”, in which operators often introduce fluctuating contact forces, positional instability, and unpredictable probing trajectories that can cause dynamic structural deformations of the arm [4]. In addition to this, the accuracy of PMCAs is dependent on robust kinematic calibration procedures that map out the geometric parameters and encoder errors of each rotary axis along the length of the arms [5, 6]. It is also important to note that the calibration procedures require the arm to be held completely still [7]. This means that the lack of a position-holding mechanism makes the system susceptible to fatigue and damage that is human in origin [1].

As a result, designing an effective joint with a position-holding mechanism is an important engineering requirement and a good practice of safety design. By designing the joint in this manner, the measurements will become independent of human error, which creates kinematic stability, improving the repeatability and accuracy of the arm.

³Articulated refers to a mechanical structure composed of a series of rigid linkages connected by rotary joints [2]

1.2 Aim and Objectives

The aim of this project is to implement a position-holding mechanism on the joint of a portable measurement coordinate arm without jeopardising its accuracy and ease of movement.

The objectives of the project include the following points:

1. To go beyond the conventional Denavit-Hartenberg (D-H) model by including force analysis due to the implementation of a position-holding mechanism on the joint.
2. To design, build, and test the joint design to be modular and applicable to a range of PCMA in industry.
3. To evaluate the effectiveness and efficiency of the joint design by verifying accuracy of measurements, through Single Point Articulation Test (SPAT).

1.3 Scope

The consideration of PCMA will be reduced to simulation, in that data sheets, user and reference manuals will be used to obtain information to model each respective PCMA for their D-H models. Only three of the most widely used PCMA in industry will be considered to influence modularity of the joint design. Only one joint will be designed, built, tested for data acquisition, through which calibration and accuracy testing will be conducted utilising the modified D-H (which includes force analysis) [7].

1.4 Motivation

PCMA are becoming more and more preferred over traditional CMMs for in-line⁴ and on-site industrial measurement [1]. This is mainly due to their portability, large working volume relative to their footprint, and their cost-effectiveness [3]. They have broad applicability across multiple industries such as aerospace, automotive, heavy manufacturing, where having a portable measurement instrument decreases handling time, production delays, and revision of work.

Despite the PCMA growing popularity, the revolute joints of these arms present an engineering challenge that has received limited investigation in literature. The joints must satisfy competing demands simultaneously:

- Tight run-out tolerances for kinematic accuracy
- Sufficient holding torque to keep the arm stationary once released
- Internal cable routing
- Compact, lightweight form that does not degrade overtime

⁴In the context of PCMA, "in-line" refers to performing quality inspections and measurements directly on the production shop floor, instead of a temperature-controlled quality laboratory [1]

This has proved difficult since there are no commercially available joints that meet these demands.

It has been established that angular position errors at each joint propagate through the kinematic chain directly affecting end-effector accuracy [5]. This proves that improvements to established joint designs have a measurable impact on overall arm performance [5]. This project addresses that gap by deriving joint specifications from a modified error model based on an error model that was developed by a former Stellenbosch University student, Leon Hall [8]. This will produce a validated joint design and testing methodology to support future PCMA development at Stellenbosch University.

1.5 Use of AI Tools

Gemini is an AI tool that will be used for to check grammatical errors and sentence structure. Claude AI will be used to assist in the formatting of the report on the online platform Overleaf using LaTeX.

1.6 Report Overview

[Give a short overview of the chapters of the report.]

2 Literature Review

PCMA's have emerged as a versatile and important tool in industrial metrology [1]. Compared to the traditional fixed Coordinate Measurement Machine (CMM), the PCMA offers portability and larger measurement volumes [1]. CMMs were originally created for inspecting the geometric paths of bent tube, allowing operators to capture 3D coordinates at the point of manufacture [3]. This has eliminated cost and delays associated with transporting component to a dedicated metrology laboratory [3]. Despite the practical advantages, the measurement accuracy of PCMA's remains a constant challenge caused by interrelating error sources.

The following reviews a few contributions to this field. First, the practical benefits and operational principles of PCMA's are put into context through industry-facing sources from Leussink Engineering and Oasis Alignment Services. The review will then discuss the academic literature covering Cuesta et al.'s virtual simulation study of dynamic deformations, Ibaraki and Saito's novel kinematic model with angular position measurement errors, Santolaria et al.'s calibration procedure based on laser tracker multilateration and lastly Brau and others' design and validation of a self-driven joint model. It must be noted that the literature reviewed for this project will be updated regularly as the project progresses.

2.1 Industry-Facing Contributions

This following papers give context and explain the contributions of each source to the project from industry.

2.1.1 The Advantages of Using a Portable CMM Arm in Metrology

This article discusses the use of PCMAAs, in particular, articulated measuring arms for accurate 3D measurement of physical objects in multiple environments such as workshops and production settings [1]. The article goes on to highlight the methodology of capturing precise measurements directly from the object to be measured [1]. This allows for real-time tracking and comparison to CAD data without moving any components. The data is transferred to CAD software, which generates 3D models and measurement reports [1]. This facilitates quality control and inspection processes. The importance of understanding the specific use of measurement data is also emphasised [1]. This can also advise interested parties when selecting the appropriate CMM arm and software platform.

2.1.2 Measurement Tool Spotlight: Portable Measuring Arms

This article discusses the evolution of PCMAAs, which were originally designed for inspecting bent tubes [3]. This highlights their long-standing presence in the study of metrology. The article emphasises the advantages of PCMAAs as opposed to traditional CMMs, specifically in terms of portability, cost-effectiveness, and efficiency in many applications [3]. It is also stated that PCMAAs are excellent tools in reverse engineering, CAD-based inspection, and first article inspection (FIA) ⁵ [3].

2.2 Academic-Facing Contributions

This following papers give context and explain the contributions of each source to the project from academic studies.

2.2.1 Dynamic Deformations in Coordinate Measuring Arms Virtual Simulation

This research paper discuss the errors induced by the structure of CMMs [4]. This highlights the significant impact of deflection caused by contact forces that occur during operation, which is noted as a key area of analysis in prior work [4]. The paper references the work of Vrhovec and Munih, who studied the deflection of CMM segments, discussing their method, which was noted to be expensive and challenging to implement [4]. There are existing standards such as VDI/VDE 2617 and ISO 10360-12 that focus on verification and calibration, however, they lack guidelines for improving accuracy and controlling probing forces [4]. It is important

⁵An FIA is a formal, comprehensive verification process that ensures a manufacturer's initial production run meets all engineering drawings, specifications, and design requirements before mass production [9]

to note that this suggests a gap in literature. The paper takes note of the fact previous studies have shown that measurement errors can reach micrometers due to unaccounted contact forces and deflection [4]. It also addresses an occurrence regarding the reliability of measurements due to human factors and the complexity of the CMM structure, leading to the non-uniform contact forces and unpredictable probing trajectories caused by operators [4].

2.2.2 Novel Kinematic Model of Articulated Arm Coordinate Measuring Machine with Angular Position Measurement Errors of Rotary Axes

This research paper indicates that measurement errors in rotary axis angular positions are mainly due to gradient errors of rotary encoder scales, which vary with angular position [5]. Previous studies have extensively used the D-H model, which includes position and orientation errors of rotary axes as error sources [5]. There is also an acknowledged challenge in ensuring sphere displacement of the stylus⁶ constrained sufficiently during SPAT due to the stiffness of the stylus holder [5]. It was noted that there exists a gap in existing models regarding the angular position measurement errors of rotary axes, which implies the need for a new or modified kinematic model. The proposed modification to the SPAT procedure aim to improve the accuracy of angular position measurements [5]. This address the limitations found in previous methodologies and a kind of framework that can be used to derive a model specific to the purposes of this project.

2.2.3 Articulate Arm Coordinate Measuring Machine Calibration by Laser Tracking Multilateration

This research paper discusses different evaluation methods for measuring arms based on established standards such as UNE-EN ISO 10360, ASME B89.4.1, and VDI/VDE 2617 [6]. This highlights their relevance in current practices in industrial metrology. The paper takes note of a gap in research, which is focused on the calibration of AACMMs [6]. This indicates that there is a lack of standardized verification and calibration procedures [6]. The paper also reveals that previous works have aimed to assess the performance of measuring arms, with some proposing methods to estimate measurement uncertainty and error sources [6]. A comparison of results with ANSI/ASME B89 volumetric performance tests is referenced, which shows good agreement and cost reduction [6]. This ultimately implies a concurrency on the effectiveness of certain evaluation methods [6]. This emphasizes the importance of multilateration⁷. techniques in improving measurement accuracy and reducing uncertainty and it reflects ongoing discussions in the metrology field regarding optimal measurement strategies [6].

⁶A stylus in a PCMA is a precision probe tip that makes physical contact with an object to capture measurement points [1]

⁷In the field of metrology, multilateration is a high-precision calibration and measurement technique that determines the 3D position of a probe or target by measuring its distance from multiple, separated, known fixed locations, normally using laser trackers [6]

2.2.4 Design and Validation of Self-Driven Joint Model for Articulation Arm Coordinate Measuring Machines

This research paper recognises that PCMA's known for their portability and cost-effectiveness, however, it is noted that they face challenges related to measurement accuracy as a result of structural parameter errors and user-induced variability [7]. Key technologies in PCMA's include kinematics modelling, error analysis, and parameter calibration, with various methods proposed for improving accuracy [7]. These methods include non-linear least squares and genetic algorithms [7]. The paper highlights the importance of temperature error compensation and the influence of external forces on measurement accuracy [7]. This indicates that there is a consensus on the need for robust error correction models [7]. Human influences, including inconsistent contact forces and measurement techniques, are identified as significant sources of error [7]. This leads to discussions on the reliability and repeatability of manual measurements [7]. The paper proposes a novel self-driven joint model to enhance measurement automation and accuracy, which addresses the limitations of traditional manual operation in PCMA's [7].

2.2.5 Error Distribution of an Articulated Arm Measurement System

This research paper discusses the lack of a universal standard for calibrating articulated arms, with the ASME standard not being widely adopted [8]. This leads to users to rely on OEM claims regarding accuracy [8]. Previous studies, such as studies by Conrad et al. and Santolaria et al., indicate that errors in articulated arms can average 0.3 mm [8], however, Renaud et al. achieved an accuracy below 1 mm using a vision-based system [8]. When these results are contrasted with the limitations of laser-based systems, which only provide positional data, it highlights a need for thorough investigations into accuracy [8]. The paper suggests that measurement errors are often not constant or linear throughout the measurement volume, which challenges OEM assertions of accuracy [8]. The paper emphasizes the importance of understanding error distribution to better plan the application of articulated arms, as errors can vary significantly across the measurement volume [8].

2.2.6 Shigley's Mechanical Engineering Design, Tenth Edition

The textbook provides different areas of relevance to the joint design of PCMA's, where stiffness, deflection control, and repeatability are the important engineering requirements [10]. Chapter 2 discusses material selection methodology, which includes the specific stiffness (stiffness-to-weight) index [10]. This supports the choice of lightweight, high-modulus materials such as aluminium alloys or carbon fibre composites for PCMA's segments [10]. This decreases gravitational deflection at extended reach positions and improves the dynamic handling characteristics of the instrument [10]. The deflection and stiffness framework of Chapter 4 is directly applicable to minimising angular and lateral compliance at each joint, as positional error in PCMA's propagate and compound along the articulated arm [10]. Chapter 7 informs the treatment of shafts and shaft components the design of the rotating elements within each joint, specifically, the management of combined bending and

torsional loads and the placement of stress concentrations away from critical cross-sections [10]. The rolling-contact bearing analysis of Chapter 11 is the main focus to joint pivot design, which covers load rating, fatigue life, and preload selection⁸[10]. Nonpermanent joint analysis in Chapter 8 remains relevant for the bolted interfaces used in general PCMA segment assembly, where controlled preload ensures repeatable joint geometry across disassembly and reassembly cycles[10].

3 Research Methodology

The project focuses on the development and implementation of a joint design with position-holding technology on a PCMA that satisfies defined metrological and performance measurements. The project, therefore, uses a design research methodology that will aid in the design, testing, and building of the joint design.

3.1 Derivation of Design Requirements

The joint specifications will be derived quantitatively from an existing error propagation model [8]. The error model relates joint run-out tolerances and encoder angular measurement to end-effector position unreliability. This will, in turn, establish measurable design requirements, including: maximum allowable run-out, holding torque, cable routing envelope, and encoder integration requirements. Data sheets, user and reference manuals from the three most popular PCMA models used in industry will also contribute to the establishment of design targets, including:

- Faro Quantum X FaroArm series
- Hexagon (formerly Romer) Absolute Arm 85 series
- Kreon Onyx measuring Arm and Skyline scanner

This will also influence the modularity of the joint design.

3.2 Concept Development

Alternative joint mechanisms are generated through literature review and mathematical reasons from first-principles. Friction hinges and damping grease such as Nyogel would provide a baseline reference, despite their inability to meet all the design requirements. Concepts will be evaluated qualitatively against the derived requirements, with infeasible options disqualified earlier on. The shortlisted concepts are refined to a comparable level of detail before a preferred joint design is chosen and justified against the requirements.

⁸Bearing preload is an important lever for eliminating backlash and improving joint stiffness while minimising excessive friction [10]

3.3 Detail Design and Analysis

The chosen joint design will be developed in full, addressing all the aforementioned design requirements. Material selection dimensional tolerances will be informed by the run-out and compactness requirements and the sources obtained from documentation of the aforementioned PCMA's most popular in industry. Analytical methods including shaft deflection calculations, holding torque modelling [10], used alongside with Finite-Element Analysis (FEA) methods such as static structural analysis will be used to verify structural and mechanical behaviour [5]. The kinematic model of the joint will be formulated using a modified D-H model convention developed by Leon Hall [8].

3.4 Design Evaluation

The final design is evaluated against the established industrially established standards such as UNE-EN ISO 10360, ASME B89.4.1, and VDI/VDE 2617 [4], and where applicable, virtual simulation. The verification will confirm that individual joint requirements have been met. Validation will assess whether the joint contributes to overall arm measurement accuracy within acceptable limits. Information obtained verification and validation and areas for further experimental validation will be acknowledged in the conclusion.

4 Workplan

4.1 Conduct Research and Review Literature

This would include looking into industry and obtaining at most, 3 of the most used PMCA models for analysis.

4.2 Investigate the Most Widely Used PCMA's in Industry

This would include doing comparative analysis across the available options, weighing pros and cons and determining whether the technology can be applied for such purposes.

4.3 Derive the Modified D-H Models

[To be completed.]

4.4 Design, Build, and Test the Joint Design

Weigh out pros and cons. Mathematical calculations and datasheets will help create benchmarks, tolerances and/or limits of performance on the joint. The best joint design will also work the best on all 3 PMCA models. Techno-economic analysis must be conducted as well.

4.5 Evaluate Results from Testing

Simulate the best joint with the three PMCAs.

4.6 Document and Finalise the Report

Analyse the kinematics and possible forces acting on the arm.

5 Resource Requirements

[To be completed.]

6 Expected Results

Briefly describe the expected results.

For PhD proposals, the unique contribution should be stated here, as well as the planned publication outputs. Publication outputs should be the same as in the Executive Summary (PG07). Title and journal/conference name should be listed. Typically 1 page or less.

7 Project Schedule

[Provide a Gantt chart here. The tasks in the Gantt chart should correspond to those of the workplan in Section ?? (Section 4). Typically 1 page.]

8 Risk Assessment

Use a risk matrix to summarize the project risks. Risk rating and mitigation should be given. Typically 1 to 2 pages.

The **Impact** Scale: Negligible (1), Low (2), Moderate (3), Significant (4), Catastrophic (5).

The **Likelihood** Scale: Very Unlikely (1), Not likely (2), Possible (3), Probable (4), Very Likely (5).

Risk Rating: Low = [1–8], Moderate = [9–16], High = [17–25]. The Impact is multiplied by the Likelihood to determine the Risk Rating. For example: Negligible (1) $\times$ Possible (3) = Low Risk (3).

Table 1: Risk Assessment

Activity	Risk	Impact	Likelihood	Risk Rating	Mitigation

9 Conclusions

[Refer to the corresponding section in the Guide for Writing Technical Reports for advice on what to include in this section. Typically 1 page or less.]

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A This is an Appendix

Note that for appendix headings use styles “Heading 1 app”, “Heading 2 app”, etc. Note that all headings must have a title, just as a chapter has a heading.

B Heading for this Appendix

B.0.1 Bla bla

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